

Software Requirements Specification

📌 Draft v0.4 / pre-baseline. This Software Requirements Specification is derived from an unbaselined Business Requirements Document. The formal Solution Architecture Document dependency remains open and is recorded as [RAID I-08](#). Publication of this page creates no architecture approval, requirements-baseline approval, implementation acceptance, UAT acceptance or release authorisation. No approval field on this page may be completed until the upstream gates are satisfied.

📖 Document structure. This page is the governing document. It holds status, purpose, scope, evidence authority, traceability, open items, readiness criteria and revision history, and it is the single entry point to the specification. The large requirement sections are held as **child pages** for size and editability. Those child pages are parts of this document, not separate artifacts: they carry no independent status, no independent approval and no independent version. The section map in section 0 is authoritative for where each section lives.

0. Section map

§	Section	Location	State
1–4	Governance, purpose, scope, evidence authority	This page	COMPLETE
5–7	System context, actors, assumptions and constraints	This page	OUTSTANDING
8	Functional system requirements	Child page	COMPLETE — SFR-001 to SFR-050
9	Non-functional and operational system requirements	Child page	COMPLETE — SNFR-001 to SNFR-020
10	Interfaces and data behaviour	Child page, planned	STAGE SRS-4

11	UI behaviour and acceptance trace	Child page, planned	STAGE SRS-5
12	Compliance and security system behaviour	Child page, planned	STAGE SRS-6 — SSEC series
13	Fully dressed use cases	Child page	COMPLETE — UC-01 to UC-09
14	UML models	Child page	COMPLETE — UML-01 to UML-09
15	Evaluation and test strategy	Child page, planned	STAGE SRS-9
16–19	Traceability, open items, readiness criteria, revision history	This page	17 to 19 maintained; 16 completed at Stage SRS-10

1. Document status and governance

Attribute	Value
Document	Software Requirements Specification (SRS)
Version	Draft v0.4 — the version applies to the whole document including its child section pages
Lifecycle state	DRAFT / PRE-BASELINE — under active authoring
Author role	Business Analyst / Systems Analyst
Content introduced	50 functional requirements (SFR-001 to SFR-050); 20 non-functional requirements (SNFR-001 to SNFR-020); 9 fully dressed use cases (UC-01 to UC-09); 9 UML models (UML-01 to UML-09). These are SRS identifiers and do not alter the BRD catalogue.

Upstream gates outstanding	Solution Architecture Document not yet formalised; BRD v1.0 baseline pending; MoSCoW Pass #1 sign-off pending
Approval record location	Change Control and Sign-Off — this document holds no approval record of its own
Governing RAID entries	I-06, I-08, I-09, I-10, I-11; risks R-01, R-02, R-05, R-06, R-07; decisions D-BRD-06, D-BRD-07; dependencies DEP-05, DEP-08

1.1 What this document does not do

- It does not alter any Business Requirements Document requirement. BR-01 to BR-06, FR-01 to FR-26, NFR-01 to NFR-16, SEC-01 to SEC-10 and UAT-01 to UAT-12 remain owned by the BRD and are unchanged by this specification.
- It does not change MoSCoW Pass #1, which remains proposed and unsigned.
- It does not record architecture approval. The Solution Architecture Document does not yet exist in the documentation lifecycle.
- It does not record UAT acceptance or release authorisation. Technical test evidence is not business acceptance.
- It does not authorise creation of a Jira project or any Jira issue. That remains gated under decision D-BRD-04.

1.2 Controlled implementation-status vocabulary

This specification uses exactly four implementation statuses, kept separate from MoSCoW priority at all times. A requirement may be a Must and simultaneously Designed / not implemented; priority describes business importance, status describes delivered reality.

Status	Meaning
Implemented	Present in the canonical repository at HEAD and supported by verification evidence
Partly implemented	Some behaviour exists and is evidenced; the complete required behaviour does not
Designed / not implemented	Design evidence exists; no implementation exists in the canonical repository

Deferred	Explicitly excluded from the current intended baseline by decision
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⚠ **Dual-status convention, pending reconciliation of [RAID I-09](#).** The catalogue pages and the Requirements Traceability Matrix record different delivery statuses for the same requirements. Until a single convention is agreed, every requirement in this specification records both the source-catalogue label and the RTM label. This specification does not silently select one.

1.3 Root cause of the status conflict

Stage SRS-2 established that the conflict recorded as I-09 is systematic rather than clerical, and that it extends beyond the requirements originally listed. The two sources are scoring different scopes of the same requirement.

- The **catalogue pages** (BRD, NFR, Compliance) score the **governed backend capability**. On that scope FR-01 to FR-05, FR-13, FR-14, NFR-01, NFR-02, NFR-06 and SEC-09 are Implemented, and repository verification supports that reading.
- The **Requirements Traceability Matrix** scores the **end-to-end user-facing capability**. On that scope the same requirements are Designed / not implemented, because no interface, API, authentication or persistent audit service exists to deliver them to a user.

Both readings are defensible and neither is wrong. The affected set is therefore larger than the entries first recorded under I-09: it additionally includes **FR-01, FR-02, FR-03, FR-04 and FR-05**. FR-13 and FR-14 are the exception — both sources agree on Implemented, because those requirements are satisfied entirely within the backend and require no user surface.

The recommended reconciliation is to adopt *Partly implemented* as the controlled status wherever a governed backend capability exists without its delivering surface, and to record the backend and surface positions as separate attributes rather than as one contested label. This remains a recommendation; the disposition belongs to I-09 and is not applied here.

2. Purpose

This specification translates the business intent recorded in the [Business Requirements Document](#) into testable system requirements, fully dressed use cases, interface and data behaviour, and UML models for the P5 Agentic RCM intelligence layer.

Its purpose is to give delivery, test and assurance roles a single controlled statement of what the system shall do, under what preconditions, producing what evidence, verified by what

acceptance method — while preserving an honest separation between what is built, what is designed, and what is deferred.

3. Scope

3.1 Product scope

P5 is a governed revenue-cycle-management intelligence layer for UAE health-insurance operations. A user asks a question in plain language; an agent selects typed tools; deterministic code queries governed views, computes compliance verdicts or produces chart specifications; a grounding validator blocks any figure that does not trace to a tool result; role checks constrain access; and human approval remains mandatory for any outbound appeal.

3.2 Intended pre-production scope

Three role-specific surfaces over one governed backend, per decision D-BRD-02: Executive Insight, Analyst Workbench and Audit Console, together with the supporting secondary screens recorded in the [UI Design Inventory and Acceptance Map](#).

3.3 Out of scope for this baseline

- Autonomous approval, autonomous outbound send, and production-write integration — deferred under decision D-BRD-05 and specified as testable exclusions SFR-047 to SFR-050.
- BRD requirements FR-23 to FR-26 — deferred; these must not become active delivery items.
- Gen-3 adaptive intake and analytical capability behind FR-19 to FR-22 — future scope, Could priority, segregated at SFR-043 to SFR-046.
- Clinical, coverage, eligibility, fraud or misconduct determination beyond approved deterministic rules, per SEC-09.

3.4 Known scope risk

The absence of an implemented user interface, application API, authentication flow and persistent audit service is a material gap against the intended three-surface scope. Twenty SFRs specify designed behaviour that currently has no delivery vehicle, and eight of the nine use cases describe behaviour delivered through a surface that does not exist. The delivery-sequencing cause is recorded as [RAID I-10](#); the component absence is recorded as risk R-05 and dependency DEP-05.

4. Referenced artifacts and evidence authority

4.1 Controlled Confluence sources

Artifact	Role in this specification	Governance state
Vision and Strategy	Strategic outcomes SO-01 to SO-08	Draft v0.1; pending stakeholder validation
Business Requirements Document	Source of all business and functional intent; requirement IDs are not altered here	Draft / pre-baseline; v1.0 baseline pending
Glossary and Data Dictionary	Controlled terminology, physical fields, semantic metrics; DD-01 to DD-14	Draft v0.1 / pre-baseline; definitions unapproved
Stakeholder Register and RACI	Actor and role authority	Pre-baseline
As-Is and To-Be Processes	Process identifiers used in requirement and use-case traces	Draft narratives; formal BPMN planned; see RAID I-07
Compliance, Privacy and Security Requirements	Business obligation source for the SSEC series at section 12	Draft; not legal advice; jurisdiction unconfirmed
Non-Functional and Operational Requirements	Business source for the SNFR series; retained as a separate supporting page	Existing; provisional SRS input
UI Design Inventory and Acceptance Map	Screen inventory and acceptance trace; retained as a separate supporting page	Design evidence only; UI not implemented

Requirements Traceability Matrix	Upstream and downstream traceability spine	Active / pre-baseline
RAID and Decision Log	Risks, assumptions, issues, decisions and dependencies referenced throughout	Active / pre-baseline
Change Control and Sign-Off	Authoritative approval and baseline register	Active / pre-baseline
UAT Plan and Scenarios	UAT-01 to UAT-12 acceptance trace; retained as a separate supporting page	Provisional / pre-SRS
Solution Architecture Document	Intended architecture authority for sections 5, 10 and 14	NOT YET FORMALISED — see RAID I-08

4.2 Evidence authority hierarchy

Where sources conflict, authority is resolved in the following order. This ordering is applied consistently throughout the specification and no conflict is resolved silently; each is recorded in section 17 or in the RAID log.

1. Canonical repository code and tests determine implementation truth.
2. Live Confluence pages determine current artifact wording.
3. Current test and evaluation evidence determines verified behaviour.
4. Architecture documents describe design intent unless code independently proves implementation.
5. User-interface images describe visual intent only.
6. Older PDF exports are historical snapshots.

7. Gen-3 documents are future design evidence and may contain implementation overstatements.

4.3 Verified implementation evidence

The following positions were verified directly against the canonical repository during pre-edit discovery on 22 August 2026 and are recorded as fact, not as claim. Every implementation status asserted anywhere in this document, including in its child section pages, derives from this table.

Evidence	Verified position
Repository HEAD	8026f57 — subject "Gen-2 close-out", dated 16 August 2026
Governed backend	Gen-1 governed backend and Gen-2 configuration-driven onboarding and portability are implemented
Registered tool contracts	Six: <code>query_claims</code> , <code>check_compliance</code> , <code>score_member</code> , <code>get_recoverable_denials</code> , <code>draft_appeal</code> , <code>render_chart</code>
Automated test suite	245 automated tests passed in the recorded isolated run
Golden set	55 items; 55 of 55 passed in the recorded dry run with zero model calls
Full live evaluation	Run 20260816T191641Z : 55 items, 49 passed, 3 automated failures (GQ-023, GQ-035, GQ-038), 3 items escalated to human adjudication (GQ-021, GQ-048, GQ-052)
Targeted live evaluation	Run 20260816T193211Z : 2 items, GQ-023 and GQ-035 passed. GQ-038 was not rerun and has not passed a live run
Post-fix live gate	INCOMPLETE — no complete 55-item post-fix live run exists; see risk R-02 and dependency DEP-08

User interface, application API, authentication, persistent audit service	Not present in the canonical repository at HEAD
Appeal capability	<code>draft_appeal</code> returns a schema-valid stub; the letter body is a placeholder. Claim particulars and filing window are real
Member scoring	<code>score_member</code> returns deterministic placeholder values and is explicitly labelled a stub; it is not P3 model output
Gen-3 capability	Adaptive intake and analytical capability are not implemented

 Technical test evidence is not business UAT acceptance. No statement in this section constitutes readiness, acceptance or approval.

5. System context and product boundaries

OUTSTANDING — narrative context and product boundary statement. The structural view is available in the meantime at [UML-02](#), which shows the user surfaces, application boundary, read-only agent plane, ingestion plane and external systems. The authoritative architecture source for this section is the Solution Architecture Document, which does not yet exist; content drafted here is subject to controlled revision under RAID I-08.

6. Actors and roles

OUTSTANDING — consolidated actor catalogue derived from the [Stakeholder Register and RACI](#). In the meantime, actors are specified per use case at [section 13](#) and shown against use cases in UML-01. Accountability for individual appeal approval remains unassigned pending decision D-BRD-07.

7. Assumptions, constraints and dependencies

OUTSTANDING — consolidated from RAID assumptions A-01 to A-06, dependencies DEP-01 to DEP-08, and the constraints recorded in the Compliance and Non-Functional pages. Per-requirement and per-use-case open items are recorded in place in sections 8, 9 and 13. No assumption is presented as an accepted service level.

8. Functional system requirements

COMPLETE — STAGE SRS-2. Held at [SRS Section 8 — Functional System Requirements](#).

SFR-001 to SFR-050. Every BRD functional requirement FR-01 to FR-26 is represented by at least one SFR, and BRD requirement text and identifiers are unaltered. Each record states a system "shall", its source BRD identifiers, process, actor or component, preconditions, inputs, required behaviour, outputs, alternative and error behaviour, security and access rule, audit and evidence requirement, acceptance method, dual implementation status, inherited MoSCoW priority, dependencies, use-case trace, UI trace, test trace and open items.

Coverage spans typed governed queries, deterministic compliance verdicts, recoverable-denials ranking, numeric grounding validation, role enforcement, chart-contract validation, the three primary surfaces, governed appeal drafting, named-human approval recording, evidence and lineage, atomic onboarding and rollback, the read-only agent-plane boundary, KPI, filter and export behaviour, the denials queue, the insights library and profile presentation. Future Gen-3 capability is segregated at SFR-043 to SFR-046 with no delivery commitment, and deferred capability is stated as testable exclusions at SFR-047 to SFR-050.

9. Non-functional and operational system requirements

COMPLETE — STAGE SRS-3. Held at [SRS Section 9 — Non-Functional and Operational System Requirements](#).

SNFR-001 to SNFR-020, derived from the [Non-Functional and Operational Requirements](#) page as the controlled business source. NFR-01 to NFR-16 are each represented and the NFR count remains 16.

 **No approved service level exists.** Of the twenty requirements, five are evidenced at the backend, seven carry a figure proposed on the NFR page but never approved, and eleven carry at least one target that does not exist at all. No numerical target has been invented. Section 9.9 holds a consolidated unapproved-target register naming each outstanding decision, its owner role and its required gate.

Catalogue coverage gap. Stage SRS-3 established that two mandated quality attributes have no requirement in the NFR catalogue at all: **portability** (SNFR-017) and **data quality** (SNFR-020). Both are recorded as derived requirements with their source marked *No source NFR* so the gap remains visible. Neither is an approved requirement until added to the NFR page under Change Control. This finding does not alter the NFR count.

10. Interfaces and data behaviour

STAGE SRS-4 — interface specifications distinguishing existing internal tool contracts, proposed application APIs, external and manual interfaces, and deferred integrations. Data behaviour derives from the [Glossary and Data Dictionary](#); DD-01 to DD-14 remain open and no meaning or classification is selected silently. Interfaces are named per use case at section 13 and shown structurally in UML-02 in the meantime.

11. UI behaviour and acceptance trace

STAGE SRS-5 — trace table across the nine screens recorded in the [UI Design Inventory and Acceptance Map](#), covering default, loading, empty, error, denied, validation and success states, keyboard operation, focus handling, contrast and non-colour cues, and the synthetic-value warning. Visual approval is not functional acceptance.

12. Compliance and security system behaviour

STAGE SRS-6 — SSEC series restating the obligations in the [Compliance, Privacy and Security Requirements](#) page as testable system behaviour with positive and negative tests. No claim of legal compliance or regulatory approval is made. D-BRD-06 remains Proposed and D-BRD-07 remains Pending.

13. Fully dressed use cases

COMPLETE — STAGE SRS-7. Held at [SRS Section 13 — Fully Dressed Use Cases](#).

UC-01 to UC-09, each specified to the full template: goal, scope, level, primary and supporting actors, stakeholders and interests, trigger, preconditions, minimal and success guarantees, main success flow, alternative flows, exception and error flows, business rules, data used, interfaces, security and privacy controls, audit and evidence generated, non-functional requirements, related UI screens, source BRD and SRS requirements, acceptance scenarios, implementation status, open assumptions and explicit exclusions.

These use cases are the behavioural spine of the specification. Every use-case identifier referenced by an SFR at section 8, an SNFR at section 9 or a diagram at section 14 now resolves to a written specification. Only UC-01 describes behaviour that exists end to end at HEAD; UC-02, UC-04 and UC-05 have implemented governed backends with no delivering surface; the remainder are designed and not implemented.

✗ UC-07 cannot be executed or accepted. The individual or role authorised to approve, reject or return an appeal draft is unassigned, because decision **D-BRD-07** remains Pending. UC-06 and UC-07 both carry the required constraints on the face of the

specification: the draft remains marked pending review, P5 cannot approve, P5 cannot send, and external submission remains outside the P5 boundary unless separately approved under a new scope package.

14. UML models

COMPLETE — STAGE SRS-8. Held at [SRS Section 14 — UML Models](#).

UML-01 to UML-09 — one use case diagram, one system context and component view, four sequence diagrams, one class model and two state machines. Each has an editable, version-controlled text source in `p5-agentic-rcm/docs/uml/` : draw.io XML for UML-01, Mermaid for UML-02 to UML-09. The source file is the authoritative artifact and any rendering is derivative. One notation caveat is recorded on that page: Mermaid provides no UML component-diagram type, so UML-02 is a structured flowchart standing in for one.

15. Evaluation and test strategy

STAGE SRS-9 — test levels, ownership, environments, data, entry and exit criteria, evidence retention, defect handling, regression, traceability and UAT hand-off. Current evidence is recorded exactly as verified in section 4.3; technical evidence is not presented as UAT acceptance.

16. Requirements traceability

STAGE SRS-10 — traceability from BRD requirement to SRS requirement, use case, UML diagram, interface and data behaviour, UI screen, test method and draft story or epic. The [Requirements Traceability Matrix](#) is updated in place rather than rebuilt.

17. Open items and unresolved decisions

The following remain unresolved and constrain what this specification may assert. None is resolved silently within this document.

Reference	Open item	Effect on this specification
I-06 / DD-01 to DD-14	Material data-definition decisions remain unresolved	Affects SFR-001, SFR-002, SFR-008, SFR-019, SFR-021, SFR-031, SFR-039, SFR-040, SNFR-020, UC-04 and UC-05;

		sections 10 and 12 remain draft
I-08	SRS drafted before Solution Architecture Document completion and BRD v1.0 baseline	The whole specification is unbaselineable; sections 5, 10 and 14 are most exposed to revision
I-09	Conflicting delivery-status labels between catalogue pages and the RTM. Section 1.3 records the root cause; the affected set additionally includes FR-01 to FR-05	Dual status is recorded on every SFR, SNFR and use case. A reconciliation is recommended in section 1.3 but not applied
I-10	Delivery sequence departed from the authoritative 2026-07 amendment; the Surfaces phase was not delivered	Twenty SFRs and eight of nine use cases specify behaviour delivered through surfaces that do not exist
I-11	GQ-048 remains unadjudicated on an apparent judge false positive	SFR-014 and the escalation acceptance scenario in UC-02 have no citable live evidence
NFR catalogue gap	Portability and data quality have no requirement in the NFR catalogue	SNFR-017 and SNFR-020 are derived and unapproved; not yet raised as a RAID item
D-BRD-06	UAE health-sector rules primary; PDPL applicability qualified	Remains Proposed; section 12 asserts no legal conclusion;

		SNFR-014 retention periods blocked
D-BRD-07	Provider versus TPA accountability for individual appeal approval and submission	UC-07 cannot be executed or accepted. SFR-022, SFR-027, SFR-029 and SFR-030 cannot name the approving role
R-02 / DEP-08	Complete post-fix live evaluation including explicit GQ-038 resolution	SFR-015 has no passing live evidence; no readiness statement may describe the live gate as green
R-06	Appeal and member-scoring stubs may be mistaken for production functions	SFR-026, SFR-050 and UC-06 record the stub boundary explicitly
DEP-05	Implemented API, UI, authentication and persistent audit service	Twenty SFRs, eight SNFRs and eight use cases specify behaviour that has no delivery vehicle
A-04 / DEP-06	Operational targets unvalidated; environment, monitoring, backup and support readiness unevidenced	Seven SNFRs carry proposed-but-unapproved figures; six carry targets that do not exist

18. SRS readiness criteria

This specification may be proposed for baseline only when all of the following hold. None is satisfied at the current draft.

- The Solution Architecture Document exists in the documentation lifecycle and sections 5, 10 and 14 have been reconciled against it.
- The BRD v1.0 baseline and MoSCoW Pass #1 sign-off are recorded in [Change Control and Sign-Off](#).
- Every SRS requirement traces upstream to a BRD requirement and downstream to a use case, an acceptance method and a test or UAT reference.
- Every use case traces to requirements and to at least one acceptance scenario.
- SATISFIED AT STAGE SRS-7, subject to the whole document remaining unbaselineable under I-08.
- Every UML diagram has an editable, source-controlled origin and a recorded validation status.
- SATISFIED AT STAGE SRS-8; validation status is Draft for all nine and no diagram has yet been reviewed.
- DD-01 to DD-14 are dispositioned to the extent that affected interface, data and acceptance behaviour is determinate.
- The delivery-status conflict recorded as I-09 is reconciled to a single controlled convention.
- D-BRD-07 is resolved so that UC-07 has a named accountable role and can be accepted.
- Every SNFR carries an approved target or an accepted, recorded deferral; no requirement enters baseline carrying an unowned TBD.
- The NFR catalogue gap for portability and data quality is closed or formally accepted.
- No unapproved numerical target is presented as an accepted service level.
- No approval, UAT acceptance or release authorisation field is populated anywhere in this document or its child section pages.

19. Revision history

Version	Date	Author role	Change
Draft v0.1	22 Aug 2026	Business Analyst / Systems Analyst	Initial framework created under Stage SRS-1. Governance panel, document status, purpose, scope, referenced artifacts and evidence authority, verified implementation evidence, open items and readiness criteria established. No BRD requirement altered; no approval recorded.
Draft v0.2	22 Aug 2026	Business Analyst /	Stage SRS-2 completed. Functional system requirements SFR-001 to SFR-

		Systems Analyst	050 authored, covering all twenty mandated behaviour areas and representing every BRD requirement FR-01 to FR-26. Section 1.3 added recording the root cause of the I-09 status conflict and identifying FR-01 to FR-05 as additionally affected.
Draft v0.3	22 Aug 2026	Business Analyst / Systems Analyst	Stage SRS-3 completed: non-functional and operational system requirements SNFR-001 to SNFR-020 authored from NFR-01 to NFR-16, with an unapproved-target register and the portability and data-quality catalogue gap recorded. Document restructured so that large sections are held as child pages under this governing document; section map added.
Draft v0.4	22 Aug 2026	Business Analyst / Systems Analyst	Stages SRS-7 and SRS-8 completed. Section 13 authored with fully dressed use cases UC-01 to UC-09, resolving every use-case identifier previously referenced by sections 8, 9 and 14. Section 14 authored with UML-01 to UML-09, each carrying an editable version-controlled source. Section 17 extended with per-use-case impact and the UC-07 accountability blocker; section 18 extended with the D-BRD-07 readiness criterion. BRD requirement text, identifiers, counts and MoSCoW Pass #1 priorities unchanged; no approval recorded.